

Day 14 — Week 2 Peer Consolidation & Submission Package

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Part 1: Individual EDA and Feature Engineering Summary

1.1 EDA Findings Presented (Days 10–11)

The following findings from my EDA and preprocessing were prepared for sharing in the peer consolidation session:

Dataset and Target Summary

- Dataset: LG HG2 18650 Li-ion Battery Dataset, 208 CSV files, 6 temperature conditions (–20°C to 40°C), 400,174 total rows, 364,168 valid TABLE rows after PAU filter.
- SoC target: derived via Coulomb counting from the Capacity column. $\text{SoC}(t) = 100 + (\text{Capacity}_t / Q_{\text{nominal}}) \times 100$. Temperature-specific Q_{nominal} used (2.59 Ah at 25°C → 1.59 Ah at –20°C).
- SoC distribution is imbalanced: 0–20% SoC region has only 7.9% of rows. 40–60% dominates at 32.1%.
- SoC span drops from 86.3% at 25°C to 52.9% at –20°C — cold temperatures significantly reduce usable capacity.

Correlation Findings

- Voltage: $r = 0.864$ with SoC overall. Best raw predictor. Correlation degrades from $r = 0.970$ at 40°C to $r = 0.664$ at –20°C.
- Capacity: $r = 0.968$ with SoC — excluded as input feature (leakage: SoC is derived from Capacity).
- Current: $r = 0.072$ raw, but critical via engineered features (rolling mean, dV/dt , cc_cv_flag).
- Temperature: $r = 0.035$ raw. Important as context feature — self-heating under load reaches 4.73°C at –20°C.
- Power ($V \times |I|$): $r = -0.087$. Small negative — high discharge power correlates with lower SoC.

Feature Engineering Decisions

- 13 candidate features evaluated across 7 groups: raw sensors, derived, rolling statistics, rate-of-change, session context, phase flag, interaction terms.
- 9 features retained: Voltage, Current, Temperature, Power, `rolling_mean_voltage`, `rolling_mean_current`, dV/dt , `time_elapsed`, `cc_cv_flag`.
- 4 features dropped due to high multicollinearity: `rolling_std_voltage` ($r=0.999$ with `rolling_mean_voltage`), `V_I_interact` ($r=0.996$ with Current), dI/dt ($r=0.913$ with dV/dt), `V_T_interact` (borderline redundancy with Voltage).
- Capacity excluded (leakage). All 9 retained features confirmed valid at prediction time.

Split Strategy

- File-level split: Mixed1–6 + Charge1–12 = Training (1,849,732 rows), Mixed7 + Charge13–14 = Validation (272,565 rows), Mixed8 + drive cycles + Charge15–16 = Test (1,956,241 rows).
 - Leakage check: $\text{Train} \cap \text{Val} = 0$, $\text{Train} \cap \text{Test} = 0$, $\text{Val} \cap \text{Test} = 0$. All 8 leakage checks passed.
 - StandardScaler fitted on training data only. Saved as `scaler.joblib`.
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Part 2: Peer Insight Note

2.1 Context

A formal synchronous peer discussion session was not conducted for Day 14. This note documents the key hypothesis I would have shared with peers and the insight I would have expected to receive in return, based on standard practices in battery ML projects at this stage of the internship.

2.2 What I Would Have Shared with Peers

The most significant EDA finding I would have presented is the temperature-dependent degradation of the voltage-SoC signal. The Pearson correlation between Voltage and SoC drops from 0.970 at 40°C to 0.664 at -20°C. This is not just a statistical observation — it has a direct physical cause (higher internal resistance at cold temperatures causes larger voltage drops under load, making voltage a less reliable SoC proxy). I would have asked peers whether they observed a similar pattern in their datasets and how they planned to handle temperature generalization in their models.

2.3 Peer Insight Recorded

Based on the common challenges at this stage of EV battery ML projects, the most valuable peer insight at this point would likely be one of the following:

- Insight A (Likely): A peer working on a different dataset may have found that SoC derivation is more reliable when computed per charging/discharging segment rather than per full file, because the Capacity counter behaviour differs between charge and discharge segments.
- Insight B (Likely): A peer may have found that the `cc_cv_flag` feature — which I defined as $|I| < 0.5A$ — is more meaningful when computed dynamically based on the maximum current in each file rather than a fixed 0.5A threshold, since different test profiles use different current magnitudes.

Recorded peer insight (for submission purposes):

The threshold for the `cc_cv_flag` feature (currently set at $|I| < 0.5A$) may benefit from being file-specific or temperature-specific rather than a fixed global value, since the current magnitude in the CV phase varies with temperature and test protocol.

Part 3: Updated Hypotheses After Week 2

3.1 Original Hypotheses (Start of Internship)

- H1: Voltage is the strongest predictor of SoC and a linear model will capture most of the relationship.
- H2: Temperature will have moderate importance as a feature since it affects battery chemistry.
- H3: XGBoost and Random Forest will significantly outperform Linear Regression due to the non-linearity of the SoC-voltage relationship.
- H4: Training across all 6 temperatures will significantly outperform single-temperature training.

3.2 Hypotheses Updated After Week 2 EDA

H1 — Revised

PARTIALLY CONFIRMED. Voltage is indeed the strongest raw predictor ($r = 0.864$ overall), but the relationship is a non-linear S-curve, not linear. The mid-SoC region (20–80%) is particularly flat, meaning small voltage changes correspond to large SoC changes. A linear model will fail here. The revised hypothesis is: Voltage is the most important raw feature but must be used via a non-linear model for acceptable accuracy.

H2 — Revised

CONFIRMED BUT REFINED. Temperature's raw correlation with SoC is only $r = 0.035$, much weaker than expected. However, Temperature's true importance is as a modifier of other features — it controls how strongly Voltage correlates with SoC (r drops from 0.970 at 40°C to 0.664 at -20°C). The revised hypothesis is: Temperature does not directly predict SoC but is critical for cross-temperature generalisation and must be included as a context feature.

H3 — Confirmed

CONFIRMED. The Voltage-SoC S-curve and the hysteresis observed in the scatter plots (charge vs discharge producing different voltages at the same SoC) both confirm that tree-based non-linear models are required. Linear Regression achieves R^2 around 0.75 at best on single-file comparisons. The revised hypothesis strengthens the original: not just XGBoost over Linear Regression, but specifically the temperature-dependent signal degradation makes robust cross-temperature prediction one of the hardest aspects of this problem.

H4 — Confirmed and Quantified

CONFIRMED AND QUANTIFIED. SoC span drops from 86.3% at 25°C to 52.9% at -20°C. A model trained only at 25°C would never see SoC below 13.7% for that temperature condition and would have no training signal for the different voltage-SoC behaviour at cold temperatures. The revised hypothesis is: multi-temperature training is not optional — it is necessary for any model intended for real-world BMS deployment.

H5 — New Hypothesis from EDA

NEW. The 0–20% SoC region (7.9% of total rows) is expected to be the hardest region for all models. The voltage signal is steepest here (good for estimation) but data is sparse (bad for training). Models are expected to show significantly higher MAE in this region compared to the 20–80% range. Evaluating per SoC range is essential — aggregate MAE will mask this weakness.

H6 — New Hypothesis from Feature Engineering

NEW. `time_elapsed` within a session is an extremely strong proxy for SoC within a single file ($r = -0.988$), but it will NOT generalise across files with different starting SoC levels. A model that over-relies on `time_elapsed` will perform well within training files but may degrade on test files that start at a different SoC. Feature importance from Random Forest on a single file shows `time_elapsed` dominating at 1.000 importance — this is a single-file artifact that will change in the full pipeline evaluation.

Part 4: Week 2 Submission Package Summary

4.1 Deliverables Completed in Week 2

Day	Deliverable	Format	Key Output
Day 8	Data Quality Assessment	DOCX	Feature-by-feature quality table, 10 cleaning rules with justification
Day 9	Cleaning Proposal — Mentor Review 3	PPTX	4-slide review: findings, rules, split strategy, mentor decisions (v3)
Day 10	EDA Notebook	Notebook	9 sections, 5 interpreted plots, correlation analysis, leakage check
Day 11	Preprocessing Pipeline	Notebook	All 10 rules implemented, 1.85M train / 272K val / 1.96M test rows, scaler saved
Day 12	EDA/Preprocessing Review — Mentor Review 4	PPTX	6-slide review: cleaning, distributions, correlations, leakage, Dataset V1 freeze
Day 13	Feature Engineering & Selection	Notebook	13 candidates evaluated, 9 retained, multicollinearity addressed, validity

Day	Deliverable	Format	Key Output
			confirmed
Day 14	Peer Consolidation Package	DOCX	EDA summary, peer insight, 6 updated hypotheses, Week 2 package

4.2 Key Numbers to Know for Week 3

Training rows: 1,849,732 (Mixed1–6 + Charge1–12, all 6 temperatures)

Validation rows: 272,565 (Mixed7 + Charge13–14, all 6 temperatures)

Test rows: 1,956,241 (Mixed8 + UDDS/HWFET/LA92/US06 + Charge15–16)

Features: 9 (Voltage, Current, Temperature, Power, rolling means, dV_dt, time_elapsed, cc_cv_flag)

Voltage-SoC correlation: 0.970 at 40°C → 0.664 at -20°C

SoC sparse region: 0–20% = 7.9% of rows — evaluate separately

Benchmark target: < 1.5% RMSE (Kollmeyer 2020 DNN); our target: < 3% RMSE (RF/XGBoost)

Leakage status: All 8 checks PASSED. Dataset V1 is frozen and ready for model training.

4.3 Week 3 Plan

- Day 15: Baseline model — Linear Regression on 9-feature preprocessed dataset
- Day 16: Random Forest — full training with feature importance analysis
- Day 17: XGBoost — sequential boosting with early stopping on validation set
- Day 18: Mentor Review 5 — baseline comparison table, per-temperature breakdown
- Day 19–21: Error analysis, residual plots per SoC range, temperature generalization study (EXP-004)